An Exact Proof of the Power-Sum Characterization in OEIS A133907

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Abstract

We prove the equivalence conjectured in OEIS A133907 between a binomial-coefficient condition and a finite power-sum congruence. Both reduce to divisibility of $\lfloor n/p \rfloor$ by p. The result is an elementary consequence of Lucas's theorem and Fermat's little theorem.

OEIS assertion. A133907, defined for $n \geq 1$ as the least prime p satisfying $\binom{n+p}{p} \equiv 1 \pmod{p}$, contains the conjecture (Thomas Ordowski, 2018) that the same least prime is characterized by $\sum_{k=1}^{n} k^{p-1} \equiv n \pmod{p}$.

Theorem 1. *For every integer $n \geq 1$ and every prime p, the following conditions are equivalent:*

$$\binom{n+p}{p} \equiv 1 \pmod{p}, \qquad \sum_{k=1}^{n} k^{p-1} \equiv n \pmod{p}, \qquad \left\lfloor \frac{n}{p} \right\rfloor \equiv 0 \pmod{p}.$$

In particular, the least prime satisfying the first condition is exactly the least prime satisfying the second.

Proof. Write $n = qp+r$ with $0 \leq r < p$. Lucas's theorem, applied to the base-p digits of $n+p = (q+1)p+r$ and p, gives

$$\binom{n+p}{p} \equiv q+1 \pmod{p}.$$

This remains valid when $q+1 \geq p$: equivalently, one may extract the coefficient of x^p from $(1+x)^{n+p}$ modulo p, using $(1+x)^p \equiv 1+x^p$, to see that the coefficient is $q+1$ modulo p.

By Fermat's little theorem, $k^{p-1} \equiv 1 \pmod{p}$ if $p \nmid k$, and $k^{p-1} \equiv 0 \pmod{p}$ if $p \mid k$ (including $p = 2$). There are exactly $q = \lfloor n/p \rfloor$ multiples of p among $1, \ldots, n$. Thus

$$\sum_{k=1}^{n} k^{p-1} \equiv n - q \pmod{p}.$$

The first congruence equals 1 precisely when $q \equiv 0 \pmod{p}$, and the second equals n under exactly the same condition. Finally, a prime $p > n$ always satisfies $q = 0$, so both least-prime definitions are well defined. □

Scope and priority. This proves the precise OEIS conjecture, but does not establish a previously unknown theorem in number theory: it follows immediately from standard classical results and the equivalent floor criterion already recorded in A133907. The contribution is a self-contained explicit verification of the posted equivalence.

References

[1] OEIS Foundation, *A133907: Least prime p such that $\binom{n+p}{p} \equiv 1 \pmod{p}$*, https://oeis.org/A133907 (accessed November 14, 2026).

[2] É. Lucas, *Théorie des fonctions numériques simplement périodiques*, American Journal of Mathematics **1** (1878), 184–196, 197–240, 289–321.

Authorship and verification. Primary author (AI): ChatGPT (OpenAI). This self-contained mathematical argument was prepared with automated arithmetic checks and source comparison. It has not received independent human peer review; no originality or priority is claimed.